

JEE MAIN REVISION MASTER: INVERSE TRIGONOMETRIC FUNCTIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 22 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Principal Value Branches, Domains & Standard Identities	22 PYQs	Inverse Trigonometry

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Principal Value Branches, Domains & Standard Identities	$\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$, $\cos^{-1}(-x) = \pi - \cos^{-1}x$, $\tan^{-1}x + \tan^{-1}y = \tan^{-1}\frac{x+y}{1-xy}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Principal Value Branches, Domains & Standard Identities (Very High Yield — 22 PYQs)

Core Principle: Principal value branches: $\sin^{-1}x \in [-\pi/2, \pi/2]$; $\cos^{-1}x \in [0, \pi]$; $\tan^{-1}x \in (-\pi/2, \pi/2)$. Negative arguments: $\sin^{-1}(-x) = -\sin^{-1}x$, $\cos^{-1}(-x) = \pi - \cos^{-1}x$, $\tan^{-1}(-x) = -\tan^{-1}x$. Fundamental co-function identities: $\sin^{-1}x + \cos^{-1}x = \pi/2$, $\tan^{-1}x + \cot^{-1}x = \pi/2$, $\sec^{-1}x + \operatorname{cosec}^{-1}x = \pi/2$.

■ **Caution / JEE Pitfall:** $\cos^{-1}(-1/2) = \pi - \pi/3 = 2\pi/3$, NOT $-\pi/3$!

■ **Speed Trick:** If $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \pi$, then $x + y + z = xyz$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): " $\tan^{-1}((1/1+1 \cdot 2)) + \tan^{-1}((1/1+2 \cdot 3)) + \dots + \tan^{-1}((1/1+n(n+1))) = \dots$ " [Ans: A]
- [2026 Shift] Q#2 (MCQ): "If $\sin^{-1}x + \sin^{-1}y = (\pi/2)$, then x^2 is equal to..." [Ans: A]
- [2025 Shift] Q#3 (MCQ): " $\sec^2(\tan^{-1}2) + \operatorname{cosec}^2(\cot^{-1}3) = \dots$ " [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

$$\tan^{-1}\left(\frac{1}{1+1 \cdot 2}\right) + \tan^{-1}\left(\frac{1}{1+2 \cdot 3}\right) + \dots + \tan^{-1}\left(\frac{1}{1+n(n+1)}\right) =$$

- (A) $\tan^{-1}\left(\frac{n}{n+2}\right)$ **[Correct]**
- (B) $\tan^{-1}\left(\frac{n+1}{n}\right)$
- (C) $\tan^{-1}\left(\frac{n}{n+1}\right)$
- (D) $\tan^{-1}\left(\frac{n+2}{n}\right)$

Correct Answer: A

Solution Analysis: Given sum is $\tan^{-1}\left(\frac{1}{1+1 \cdot 2}\right) + \tan^{-1}\left(\frac{1}{1+2 \cdot 3}\right) + \dots + \tan^{-1}\left(\frac{1}{1+n(n+1)}\right)$ The general term is $\tan^{-1}\left(\frac{1}{1+k(k+1)}\right)$ Using the identity $\tan^{-1}x - \tan^{-1}y = \tan^{-1}\left(\frac{x-y}{1+xy}\right)$ take $x=k+1$ and $y=k$. Then, $\tan^{-1}(k+1) - \tan^{-1}k = \tan^{-1}\left(\frac{(k+1)-k}{1+k(k+1)}\right) = \tan^{-1}\left(\frac{1}{1+k(k+1)}\right)$ So each term can be written as $\tan^{-1}\left(\frac{1}{1+k(k+1)}\right) = \tan^{-1}(k+1) - \tan^{-1}k$ Now the sum becomes $[\tan^{-1}2 - \tan^{-1}1] + [\tan^{-1}3 - \tan^{-1}2] + \dots + [\tan^{-1}(n+1) - \tan^{-1}n]$ All middle terms cancel, leaving $\tan^{-1}(n+1) - \tan^{-1}1$